"Bridging Cognitive Processing and Social Dynamics New perspectives or

30:42

welcome
back welcome back hello son and Ma again
we are in the session physics and
metaphysics of social Powers so thank
you both for joining and take it
away thank you so much for having us I
was just asking if we should introduce
ourselves which uh which we
will um so I will introduce myself and
then ma can introduce herself so that I
don't make any mistakes so because my
brain is on
hyper hyper abstractions at the moment
anyway I am Sonia I'm a PhD student uh
based in Rotterdam at the araso school
of
philosophy and uh we've been working on
this paper with ma recently and Ma
please uh yes so I'm a PhD candidate at
the University du amoral in cognitive
Computing and I'm also a director of
research strategy at versus
thank you so yeah as Daniel said we're
going to be talking about the the
Spectrum the big spectrum of between the
physics and the metaphysics of uh social
Powers uh very inspired by the work of
uh of ramstead here HP Hines and uh by
Mouse work on on on power recently with
Sarah Grace mansky um and then we're
trying to combine that with some work
that I'm doing with two other colleagues
uh Wilkins and clav Vasquez on on the
concept of active ignorance which we
will be um continue continuing to work

on in the coming
time um so yeah and then generally
inspired by by ideas of semantic
attraction from Hinton and by Major
names in in philosophy and sociology who
have been treating these Concepts
through different Avenues uh less kind
of formal Avenues but still relevant
nonetheless now that we're actually
capable of joining um yeah the physical
with physical that's what we're going to
we're going to try here with this
presentation so the concept of power we
we argue can be explored at at several
scales from physical action and process
effectuation all the way to complex
social dynamics so this kind of spectrum
wide analysis of power requires
attention to the fundamental principles
that constrain these
processes and in the social realm the
acquisition and maintenance of power is
intertwined with both social interaction
and cognitive processing
capacities so socially facilitated
empowerment grants empowered agents more
information processing capacities and
opportunities either because they're
relying on others to bring about desired
policies and or because they're able to
enjoy more information processing
possibilities as a result of relying on
others for the reproduction of material
tasks material is in Brackets because
these tasks need not be material so the
the effects of social empow empowerment
imply an increased ability to compute
information thereby

augmenting the evolution of a specific
State space specific is a keyword here
uh really commenting back on what Ma
just uh just mentioned in at the end of
the previous
talk so empowered individuals attract
the attention of others attention is one
the central Focus uh of of our paper so
they attract the attention of others who
contribute to increasing the scale of
their access to various policies
effectuating these these State spaces
that we mentioned so the presented
argument posits that social power in the
context of active inference is a
function of several valuable gr
variables among them an individuals or
group's ability to attract the attention
of others with access to or the ability
to facilitate desired
policies and an individuals or group
capacity to effectively process
information as a result of its power
amplifying effects this extended
computational ability also buffers
against potential loss or or possible
vulner
vulnerabilities semantic social
artifacts so narratives um ideologies
histories representation at large are
attentional scripts which ensure the
coordination of social patterns so you
can think of anything from psychological
schema habits group think insulation
policies Etc and these are the
structures under and by which agents
mediate and leverage power and they're
because of that the main focus of

attention
here so uh thinking about this the
linking between the physical and
metaphysical Concepts and the world
gravitational metaphors abound in
language this is like really prominent
in the work of of Shephard Lake of and
Johnson of course with spatial metaphors
um I've been I've done some writing on
it at the beginning of my PhD and also
um laand Kent uh did some really nice
work that's also up on the on the active
inference archive and then in his papers
course um so the idea of power being
pulled towards complex social attractors
uh was also famously presented by uh
Pier borue who linked various concepts
of capital cultural capital political
cap Capital Social Capital to power as
this kind of social attraction and also
traction and he borrowed the metaphor
from from physical field theory in fact
so there is already the connection is
really present and obvious there uh only
in in a metaphoric way um which we're
kind of trying to expand on here uh so
expanding also on his legacy to to you
know take the epistemological and the
empirical as totally Inseparable things
that we're analyzing and his desire to
actually ground sociological analysis in
in in a scientifically tractable manner
we're trying to frame uh these kind of
attentional attractors that move systems
towards certain configurations through
active
imprints and this framing allows for a
new take on the dialectics between the

individual and the collective again
linking back to what was just mentioned
at the end of the previous talk so in
terms of um questions of information
processing capacity not only are
resource distrib ution and you know the
division of labor a systemic fact but
also the modulation and projection of
future possibilities cannot be computed
at the level of the individual clearly
so this is something rather obvious in
our context and our framing power this
is how and why the minimization of
uncertainty for for social systems is in
fact social right so agents seek
alliances and policies which ensure that
cognition so information processing and
future projection can be distributed
this allows for efficient practical
traction because and here I quote from
Steinberg at all 2023 epistemic
confidence in knowing interpreting and
acting together is that traction that
happens when we when we're socially
distributed uh and this in contracts
with this kind of individualized agency
where one's estimation of and here I
quote again Divergence in mental States
between self and others is what allows
for the sort of prediction and
interpretation of others Behavior which
we might kind of want to frame as
something that has a shorter range
traction on on the future so now we're
going to look at the theoretical
framework and I pass it over to
Ma so um obviously you have heard this
15 times but I'm still gonna go through

it because there are some Concepts in here that will become relatively silent after but to go over it briefly active inference explains how biological agents minimize varal free energy to align their internal beliefs with external reality so effectively they've reduced discrepancies between expected and actual sensory inputs and we sort of Leverage The Descent on free energy to to to couch model uh perception cognition and action through basian uh belief updating and so free energy is the upper bound on surprise or the unexpectedness of sensory data relative to an agent's predictions and so direct computation of surprise is complex so agents use variational free energy uh to to approximate based on their internal in generative model and so um you can see that for example the K Divergence term measures how closely the internal uh beliefs match the true posterior of and so agents align their beliefs with actual data as they minimize it so the log likelihood term ensures predictions align with observations so to minimize fre energy agents update their beliefs they allow for adaptive action selection that supports coherent gold directed behavior and so there are two strategies perceptual inference to adjust your beliefs to match the sensory inputs and active inference which is um acting to change the environments and align new sensory data with internal
models um active inference interestingly and as we saw earlier with the the whole

word cloud spans or can span multiple
scales from cellular processes
maintaining homeostasis through
cognitive systems using predictive
coding to update beliefs based on
hierarchical prediction errors and so
this really gives us a broad biological
principle where minimizing free energy
promotes systemic efficiency and optimal
homeostasis across scales and so um
reducing free energy really means that
you minimize entropy as well and you
align your beliefs with reality to
reduce this
uncertainty important to not is that we
also Factor um action and so expected
free energy can be decomposed to
extrinsic goal oriented and epistemic
uncertainty reducing components that
balance exploration and exploitations
and so this dual strategy ins ensures
that agents achieve immediate goals
while gathering information to adapt
effectively in uncertain
environments and so Central to active
inference we really have this concept of
a generative model which allows agents
to predict sensory inputs and their
hidden causes the generative process
represents the real world states that
produce observations and generative
models reflect the agent's hypotheses
about these processes uh the agent's
actions correct mismatches between its
model on real sensory inputs uh to align
the model with reality going quickly
over uh the the different terms in a
hierarchical structure each level

predicts the states of the ones below and it corrects predictions based on sensory feedback and there are conditional depend encies that Cascade between layers that support uh belief updating and
adaptation we have observable states that are the sensory inputs the agent receives and tries to predict using its model the hidden States uh latent variables that represent underlying causes inferred by the agent to explain sensory inputs there is a series of parameters that Define the statistical dependencies between hidden and observable States likelihood function which indicates how well the hidden States and parameters predict observe data we have a PR distribution which is the initial beliefs about the hidden States and the posterior distribution which is the updated beliefs after observing data and combining likelihood and prior then obviously we use the marginal likelihood which is the probability of observed data under the model to integrate over hidden States and so we use this for model evaluation and comparison to determine how well a model explains sensory inputs so the generative model really enables the agents to align its beliefs with the sensory data sometimes through hierarchical predictions realistically most often through hierarchical predictions and feedback loops and they refine their understanding of the environments to improve action selection

for adaptive behavior and so this
picture is really uh very
individual so
um we can see though that within this
picture it's really really Rife to uh to
couch power dynamics so we start from
Individual agents as predict systems
that minimize uncertainty through
prediction uh through perception and
action and so brunenberg really made a
Salient point about this with his paper
on the Primacy of I can where agents
develop you know empowerment through
embodied interactions which enable them
to affect the
environment so if we scale this up to
Social Power agents use generative
models for self- prediction but also to
predict others intentions they form the
basis social understanding in Collective
action Carl has lots of work about this
um Connor Hines as well Axel constant
maxel ramstead Casper hes Natalie
Castell which I believe will give a talk
as well and so
attention uh is driven by Deep
preferences and it shapes what becomes
Salient which influences both individual
and group Behavior at the group level
shared predictive models
facilitate coordinated actions so
stigmergic processes as explored by our
very own Daniel uh environmental
modifications that guide Behavior allow
distributed control while deontic cues
as explored by axle constant enforce
social norms and so cultural practices
further shape Collective attention it

creates shared perceptual Landscapes
which is something that we explored with
uh Toby Sinclair Smith I think I said
his name wrong I call him Toby so I
don't really know how to say his name
and finally we have to know um how to
couch joint attention and this is work
that and at the University of Quebec in
Montreal have also positive that posited
that this mechanism basically enables
efficient coordination through mutual
recognition of action
possibilities and
so power here our structures emerge
through self organizing
subgroups based on belief similarity so
again this is work that has been
explored by Natalie
Castell which we can call
decentralized influenced spreads within
Network structures and they amplify
certain narratives While marginalizing
others and so these these processes
create cultural C Capital where shared
norms and knowledge facilitate efficient
social
interaction
um attention here is thus shaped by
those preferences but it shows that the
importance uh of critically constructing
shared narratives without homogenizing
scripts that stifle diversities are is
an important thing to consider so it's a
point made by Anna AR simplistic one
dimensional fuse like there is no
alternative constrain attention and
reduce complex systems to narrow goals
which is why fascism tends to be very

attractive especially in times of high
uncertainty and so distributed
generative models over time including
historical narratives and generational
Dynamics give us the tools to Envision
power as possibilistic which is a term
that Sonia introduced here this this
shifts the focus from isolated agency to
Collective Dynamics and hierarchical
organization we get a richer analysis of
how agents and groups interact within
con constraints to shape future
possibilities so active inference really
gives us a frame for social power beyond
very simplistic atomized ideas of agency
and focus instead on this emergent
distributed process that structure
Collective behavior and
potential thank
[Music]
you
so yeah to return to the concept of
narrative because it's sort of the thing
that that um frames something that might
go from from from the small scale to the
large scale in different ways and
something that we can all colloquially
uh sort of latch onto narratives are are
are social mechanisms which have the
power to make or break power you can say
uh the cognitive exploration of of of
the not yet engaged right of of scripts
that are not yet there but one may kind
of uh uh latch onto or sort of
effectuate anything you can think of
here things like speculations fictions
future possibilities of All Sorts um can
be these can be under as semantic

attractors that enable kind of relatively risk-free learning right so you're you're projecting to imagine what might be the case if and the simpler The Narrative usually the easier it is to sort of minimize like vast uncertainty so these possibilistic Explorations in the form of narratives allow agents to explore counterfactual scenarios and minimize uncertainty without the sort of exposure to the to the real world consequences of this um this this SA safy mechanism frames both The evolutionary advantage and persistent appeal of narrative engagement in human culture and in the context of power you know leveraging strong narratives as as Ma just pointed out such as US versus them that's like the the the one that you know the biggest semantic attractor that we seem to keep returning to all the time these types of reductionisms that greatly minimizes uncertainty about an agent or group's current predicaments and they become because of this very very appealing and sometimes self-fulfilling unfortunately policies that bring about their effects resulting in Collective uh Collective behaviors which can benefit sorry I'm so like which can be beneficial or detrimental to the greater whole the uh the effects of of variational free energy minimization can be observed of course at the level of the basic structures of language in this sense so semantic attractors like redundancy rhyme alliteration Etc can be

stood as patterns which Aid memory
communication informational retrieval
policies because they permanently
reappear right so pulling attention to
the same phenomena and here I use scare
quotes very very strongly the same
phenomena uh thus greatly reducing the
need to interpret and organized
linguistic interaction and cognition a
new every time which is very
costly so we can understand for example
the possible emergence of rythmic
patterning in communication as enabling
the storage and access of information
collectively by connecting a sound
pattern to a phenomenon we reduce
uncertainty about reality once that
sound pattern is witnessed again in in
sort of true pavlovian
fashion by way of the same uncertainty
minimizing mechanisms simple simple or
complex narrative constructs create
cognitive situations where agents reduce
comple complex aspects of the world down
to Preferred
scenarios and now we can look at how
active inferences are useful apply
framework for this I pass it over to
again thank
you so I'm I'm here drawing on work that
is currently ongoing from other people
as well with whom I'm collaborating and
I want to make sure that these ideas are
appropriately attributed to
them we're going to talk about
individual agency and empowerment and so
autonomy uh can be defined as a system's
capacity to self-govern using internal

Norms uh so this is a topic explored by
Jaclyn Hines Alex kefir inish poo in
part ADV veres um and so empowerment
here is measured by the maximum Mutual
information between an agent's actions
and resulting States so this reflects
how much influence an agent has over
future States uh higher empowerment
means greater potential for desired
outcomes and so empowerment maximizes is
the entropy of actions and States while
ensuring strong action State
correlations autonomous agents are thus
more driven by intrinsic motivation and
this is a premise that again Alex Kefir
is one of the Pioneers uh
here it's thus connected to drives and
control to systems that optimize future
action State paths and so under active
inference agents minimize expected free
energy to balance exploration
information seeking and exploitation
goal Pursuit so this involves perceptual
inference to update beliefs and generate
model refinements through action
outcomes and so here power operates as
an attractor in Social systems it it
draws agents towards influential
entities through access to resources and
strategic advantages Happ it leads
through it leads therefore to to
amplification through feedback loops um
powerful agents help others minimize
their expected free energy by providing
some degree of stable predictable
Frameworks and reshaping the
informational
landscape so high power entities have

privileged access to information and can
model their environment more effectively
and they also influence what can be
modeled and how impacting the collective
predictive landscape
powerful agents can offload cognitive
demands as well processing complex
information more efficiently and
directing narratives to manage error
absorption and uncertainty so power here
enables agents to shape others Focus
directing attention to favorable States
while diminishing access to Alternatives
and this leads to feedback loop where
control over the environment enhances in
fluence and perpetuates power the whole
story around
1% so power grants a broader range of
operational capabilities and extends
influence over others State space
policies within its domain and so here
active inference reveals power as this
emerging property um of the connection
between perception action and social
structures more powerful agents maintain
and expand their control by shaping both
informational and physical
environments so let's go over three main
components Precision modulation belief
alignment and ultimately minimizing
Divergence first Precision
modulation powerful agents shape
expected outcomes by altering the social
or informational environment as we saw
they create narratives or scripts that
effectively reduce the dimensionality of
the state space forming stable
attractors such as social norms or

shared stories and these attractors
direct Collective attention and ensure
that other agents align their actions
with these preferred
States um but here basically the
posterior belief about States uh has uh
an alpha that modulates the impact of
precision on these beliefs and so in
belief alignments the power to direct
attention and utilize Precision
modulation really enables the leaders to
make specific aspects of reality more
Salient for others and so by
establishing these attractors within the
belief space the leaders guide others
interpretations and behaviors um these
attractors act as a focal points
reducing ambiguity and encouraging
agents to align their beliefs and
actions around them thus fostering a
shared understanding so the the
alignment process naturally emerges for
minimizing the expected free
energy uh and so for example um uh where
P of O given s and Pi is the likelihood
of outcomes uh given States under policy
pi and uh Q of s given o and Pi is the
posterior belief about States and so
this decomposition shows us the
intrinsic and extrinsic components which
balance uncertainty reduction with goal
oriented behavior that is um shaped by
this Precision that we mentioned earlier
and so finally uh minimizing Divergence
uh can be represented just using a
coolback Liber Divergence uh which is a
measure of the difference between two
agents belief distributions and so a

lower K Divergence indicates closer
belief alignments and so leaders
influence this alignment by modulating
this Precision which amplifies
confidence in specific interpretation
and so the process minimizes belief
Divergence across agents which promotes
synchronized understanding and cohesive
group Behavior without the necessity for
direct coercion although that is yet
another way uh to minimize the state
space uh and therefore is an available
practice so Precision modulation sets
the framework for control belief
alignments ensures Collective cohesion
and minimizing Divergence uh formalized
as the alignment so basically we capture
the Dynamics uh and influence uh the
Dynamics dynamics of power and influence
within social
systems and so finally we come to the
conclusion we have a lot more in the
paper that we're still working out so
look out for
that um yeah so what we're we're
thinking about here in terms of this
kind of shift um attentional shift uh
from from ideas of territorial and
material power to this new realm of of
informational and computational power um
well this this for us requires attention
to attention uh for sure something that
we can we can look at in uh in ways that
we can formalize and of course we
propose that active inference provides
crucial theoretical tools for for making
power structures visible and therefore
analyzable and so allowing possibly for

the potential to rethink current social
asymmetries so again given this of
apparent transition between the
organization of life around the
centrality of physical work and matter
and things moving in space towards the
centrality of attention and
information uh especially considering
the whole attention economy
idea we kind of try to think about the
possible sustainable social future uh
which requires the addressing of both
resource and information distributions
and the latter is best analyzed through
the power of attention um as we've tried
to to show as a major factor or the one
major Vector influencing social
dynamics and of course this is very
important to mention hierarchies are
inevitable in complex thermodynamic
systems but we propose if we can begin
thinking about this uh fostering
transparent plastic hierarchies that
remain contestable and thus open to
novelty open to learning right this
approach um has to acknowledge uh
historic and thermodynamic constraints
while also acknowledging space for
systemic
Evolution and while hierarchies are
inevitable social scripts should be read
writable by those with an interest in
survival within them right and this
helps reorient our understanding of an
unfolding globalizing interconnected and
interdependent life and participation in
a system is only participatory if those
involved are able to track the

consequences of the variables and th
future State spaces at stake a true
distribution of information processing
cap capabilities is essential for
systemic sustainability and proper
Equitable development and when thinking
through evolutionary biology lenses for
example let's try to polish them clearly
and not fall into social Darwinism
narratives and to quote from the xenop
feminist
Manifesto um lovely colleagues there
Patricia Reed for example if nature is
unjust then we change nature which is
what we are all in the business of doing
here at little
bit all right that was um that was our
presentation thank you
awesome okay we'll be
back in a few
minutes
e for